

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

C04-ASSESSMENT TOOL 3 – Viva Based Questions

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 – Viva Based Questions	Assessment:	Assessment Tool 3– Viva Based Questions
Weightage:	20% of CIA	Total Marks:	100
CO4 Statement:	To apply the statistical inferences such as testing of hypothesis and point estimates for the given data		
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Section / Batch:	2024	Date:	21-08-2026

Code of Conduct

I _ T.Chandana (192472059)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _ T.Chandana _____

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

A. INTRODUCTION TO STATISTICAL INFERENCE

1.What is statistical inference?

Statistical inference is the process of using information from a sample to make conclusions or predictions about a larger population.

2.Why is statistical inference important in data science?

Statistical inference helps data scientists make decisions about a population using sample data. It is useful for estimation, prediction, hypothesis testing, and decision-making under uncertainty.

3.What is the difference between descriptive statistics and inferential statistics?

Descriptive statistics summarizes and describes the data collected, such as mean, median, and standard deviation. Inferential statistics uses sample data to make conclusions or predictions about a population.

4.What is a population in statistics?

A population is the complete set of individuals, objects, or observations that a statistical study is interested in.

Example: All students in a university.

5.What is a sample?

A sample is a smaller group selected from a population for analysis.

Example: 100 students selected from a university containing 5,000 students.

6.Why do data scientists usually work with samples instead of complete populations?

Data scientists usually work with samples because studying the entire population may be expensive, time-consuming, or difficult. A properly selected sample can provide useful information about the population.

7.What is a population parameter?

A population parameter is a numerical value that describes a characteristic of the entire population.

Examples: Population mean (μ), population proportion (p), and population variance (σ^2).

8.What is a sample statistic?

A sample statistic is a numerical value calculated from sample data.

Examples: Sample mean ($\bar{x}$), sample proportion ($\hat{p}$), and sample variance (s^2).

9.Give a real-world example where statistical inference is used in industry.

A company may survey 500 customers to estimate the satisfaction level of all its customers. The results from the sample are used to make conclusions about the entire customer population.

10.What are the two major activities involved in statistical inference?

The two major activities are:

1. Estimation – estimating population parameters from sample data.
2. Hypothesis testing – testing claims or assumptions about a population.

B. FREQUENTIST APPROACH**11.What is the frequentist approach to statistical inference?**

The frequentist approach is a method of statistical inference where probabilities are interpreted using long-run frequencies of repeated experiments.

12.What is the basic idea behind frequentist statistics?

The basic idea is to use sample data to estimate or test fixed population parameters and make decisions based on how frequently results would occur in repeated sampling.

13.How does the frequentist approach differ from the Bayesian approach?

In the frequentist approach, population parameters are treated as fixed but unknown values. In the Bayesian approach, parameters are treated as uncertain quantities and probability distributions are used to represent beliefs about them.

14. In the frequentist approach, what is considered fixed: the parameter or the data?

The population parameter is considered fixed but unknown, while the observed data is considered random because it can vary from sample to sample.

15. What does long-run frequency mean in frequentist statistics?

Long-run frequency means the proportion of times an event occurs when an experiment is repeated a very large number of times under the same conditions.

16. Why is random sampling important in the frequentist approach?

Random sampling helps ensure that each member of the population has a known chance of being selected. It reduces selection bias and allows valid statistical inference about the population.

17. Can a population parameter be considered a random variable in the frequentist approach? Why?

No. In the frequentist approach, a population parameter is considered a fixed but unknown value. The randomness comes from the sample and the data collected.

18. Give an example of a frequentist problem in a data science application.

A data scientist may test whether a new machine-learning model has a higher accuracy than an existing model. Sample results are used to perform a hypothesis test and determine whether the observed improvement is statistically significant.

C. POINT ESTIMATION

19. What is a point estimate?

A point estimate is a single numerical value calculated from sample data and used to estimate an unknown population parameter.

Example: The sample mean $\bar{x}$ is a point estimate of the population mean μ .

20.What is an estimator?

An estimator is a statistical rule or formula used to estimate a population parameter from sample data.

Example: The sample mean is an estimator of the population mean.

21.What is the difference between an estimator and an estimate?

An estimator is the formula or rule used to estimate a parameter. An estimate is the numerical value obtained after applying the estimator to a particular sample.

Example: $\bar{x}$ is the estimator, while $\bar{x} = 72$ is the estimate.

22.What is the point estimator for the population mean?

The sample mean is the point estimator for the population mean.

Formula:

$$\bar{x} = \Sigma x / n$$

23.How do you estimate a population proportion from a sample?

The population proportion is estimated using the sample proportion.

Formula:

$$\hat{p} = x / n$$

where x is the number of successes and n is the sample size.

24.What is the point estimate of population variance?

The sample variance is commonly used as the point estimate of population variance.

Formula:

$$s^2 = \Sigma(x - \bar{x})^2 / (n - 1)$$

25.Why is the sample mean commonly used to estimate the population mean?

The sample mean is commonly used because it is an unbiased estimator of the population mean and generally provides a reliable estimate when the sample is representative.

26.What properties should a good estimator have?

A good estimator should ideally have:

1. Unbiasedness
2. Consistency
3. Efficiency
4. Reliability

27.What is an unbiased estimator?

An unbiased estimator is an estimator whose expected value is equal to the true population parameter.

For example:

$$E(\bar{x}) = \mu$$

Therefore, the sample mean is an unbiased estimator of the population mean.

28.What is meant by the bias of an estimator?

Bias is the difference between the expected value of an estimator and the true population parameter.

Formula:

$$\text{Bias} = E(\text{Estimator}) - \text{Parameter}$$

An estimator with zero bias is called an unbiased estimator.

29.What is the difference between bias and variance of an estimator?

Bias measures how far the expected value of an estimator is from the true parameter. Variance measures how much the estimator varies from sample to sample.

Bias represents systematic error, while variance represents sampling variability.

30.Why can two different samples produce different point estimates for the same population parameter?

Different samples may contain different observations. Therefore, their sample statistics, such as sample means or proportions, can be different even though they come from the same population.

D. VARIABILITY IN ESTIMATES

31. Why does a point estimate have uncertainty?

A point estimate is calculated from a sample rather than the entire population. Since different samples can produce different estimates, the estimate has uncertainty.

32. What is the sampling distribution of an estimator?

The sampling distribution is the probability distribution of an estimator obtained by considering all possible samples of the same size from a population.

33. What is the standard error?

Standard error is a measure of the variability of a sample statistic from one sample to another.

For the sample mean:

$$SE = \sigma / \sqrt{n}$$

When population standard deviation is unknown, the sample standard deviation is commonly used:

$$SE = s / \sqrt{n}$$

34. How is standard error different from standard deviation?

Standard deviation measures the variability of individual observations within a dataset.

Standard error measures the variability of a sample statistic, such as the sample mean, across different samples.

35. How does increasing the sample size affect the standard error of the sample mean?

Increasing the sample size decreases the standard error.

Formula:

$$SE = \sigma / \sqrt{n}$$

As n increases, $\sqrt{n}$ increases, so the standard error decreases.

36. Why does variability in an estimate decrease when the sample size increases?

A larger sample contains more information about the population. Therefore, random sampling fluctuations have less effect on the estimate, resulting in lower variability.

37. What is the relationship between standard error and confidence intervals?

Standard error determines the uncertainty around an estimate and directly affects the width of a confidence interval.

General formula:

Confidence Interval = Estimate $\pm$ Critical Value $\times$ Standard Error

A larger standard error produces a wider confidence interval.

38. If two samples have different sample means, does it necessarily mean that one sample is incorrect? Explain.

No. Different sample means are normal because samples contain different observations. Sampling variability can cause different samples from the same population to produce different means.

E. CONFIDENCE INTERVALS

39. What is a confidence interval?

A confidence interval is a range of values calculated from sample data that is used to estimate an unknown population parameter at a specified confidence level.

Example: A 95% confidence interval for a population mean may be (50, 60).

40. Why is a confidence interval more informative than a point estimate?

A point estimate gives only one value, while a confidence interval provides a range of plausible values and shows the uncertainty associated with the estimate.

41. What does a 95% confidence level mean in the frequentist approach?

A 95% confidence level means that if we repeatedly take samples and construct confidence intervals using the same method, approximately 95% of those intervals would contain the true population parameter.

42.What is the difference between a confidence level and a confidence interval?

Confidence level is the percentage of confidence associated with the estimation procedure, such as 90%, 95%, or 99%.

Confidence interval is the actual numerical range calculated from the sample data.

Example:

95% = Confidence level

(45, 55) = Confidence interval

43.What factors determine the width of a confidence interval?

The main factors are:

1. Sample size
2. Variability or standard error
3. Confidence level
4. Critical value

44.How does increasing the sample size affect the width of a confidence interval?

Increasing the sample size decreases the standard error. Therefore, the confidence interval becomes narrower and the estimate becomes more precise.

45.How does increasing the confidence level from 95% to 99% affect the confidence interval?

Increasing the confidence level from 95% to 99% increases the critical value. Therefore, the confidence interval becomes wider.

46.What is the difference between a z-confidence interval and a t-confidence interval?

A z-confidence interval uses the normal distribution and is commonly used when the population standard deviation is known or when the sample size is sufficiently large.

A t-confidence interval uses the t-distribution and is commonly used when the population standard deviation is unknown, especially for smaller samples.

47. When would you use the t-distribution instead of the normal distribution for estimating a population mean?

The t-distribution is generally used when the population standard deviation is unknown and the sample size is small, particularly when the population is approximately normally distributed.

F. HYPOTHESIS TESTING USING CONFIDENCE INTERVALS

48. What is hypothesis testing?

Hypothesis testing is a statistical method used to evaluate a claim about a population using sample data.

It involves comparing the evidence from the sample with a stated hypothesis and making a decision.

49. What are the null hypothesis (H0) and alternative hypothesis (H1)?

The null hypothesis (H0) is the initial statement or claim that there is no significant difference, effect, or relationship.

The alternative hypothesis (H1) is the statement that contradicts the null hypothesis and represents the presence of a significant difference, effect, or relationship.

Example:

H0: $\mu = 50$,

H1: $\mu \neq 50$

50. How can a confidence interval be used to make a decision about a hypothesis?

A confidence interval can be compared with the hypothesized parameter value.

If the hypothesized value falls inside the confidence interval, we generally fail to reject H0.

If the hypothesized value falls outside the confidence interval, we generally reject H0 at the corresponding significance level.

Example:

Claim: $\mu = 50$

95% Confidence Interval = (52, 58)

Since 50 is outside the interval, the null hypothesis is rejected.

Final concept:

Confidence intervals help us estimate the population parameter and also provide a way to make decisions about statistical hypotheses.